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SENIOR SECONDARY SCHOOL, TRICHY-620 008
GRADE XII - PRACTICAL PROGRAMS – 2026-27

GENERAL INSTRUCTIONS IN WRITING THE COMPUTER SCIENCE RECORD

1. **Questions and Outputs should be written in the Left hand "Unruled" side in Pencil.**
2. **Aim, Coding and Result should be written in the Right hand "Ruled" side in Ink Pen.**
3. **Question is available at the top of every question's coding.**
4. **Leave TWO lines BLANK at the top of right side for every question to fill up the heading later.**
5. **When we add the word "To" in front of every question, it becomes Aim.**
6. **Please take utmost care in writing the CODING by providing indentation wherever it is necessary.**
7. **The Result for all the Programs is common which was typed in the next page.**
8. **Normally the Outputs should be drawn in the left hand "Unruled" side of the page where Result has been written. But if we have many outputs to be drawn, we can start using the previous left hand "Unruled side itself**
9. **Give importance to Presentation by drawing the border in the left hand side as it will project our qualities as well.**
10. **Don’t write the Bonafide certificate Page, Index page and Page numbers inside the Record note now**

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Question No :

Write the question exactly as it is after the question no

For example:

Question No : 1

Write a program to find whether the entered number is Perfect or not

Outputs:

Blank for time being
Aim :
To Write a program to find whether the entered number is Perfect or not
Coding:
Write exactly as it is given in the PDF.
Care must be taken in providing the Indentation properly
Result :
○ **The Program was successfully completed**
○ **The Output was verified by giving sample values**

---

1. Write a program to find whether the entered number is Perfect or not

```
num=int(input("Enter a number "))
divsum=0
for i in range(1,num):
    if num%i==0:
            divsum=divsum+i
if num==divsum:
    print(num,"is a Perfect number")
else:
    print(num,"is NOT a Perfect number")
```

---

2. Write a program to find whether the entered number is Armstrong or not

```
num=int(input("Enter a number "))
digits=len(str(num))
sum=0
cnum=num
while num>0:
    sum=sum+(num%10)**digits
    num=num//10
if cnum==sum:
    print(cnum,"is a Armstrong number")
else:
    print(cnum,"is NOT a Armstrong number")
```

---

3. Write a program to find whether the entered is Palindrome or not

```
num=int(input("Enter a number "))
revnum=0 ; cnum=num
while num>0:
    revnum=revnum*10 + num%10
    num=num//10
if cnum==revnum:
            print(cnum,"is a Palindrome")
else:
            print(cnum,"is NOT a Palindrome")
```

---

4. Write a program to find whether the entered number is Prime or not

```
num=int(input("Enter a number "))
for i in range(2,num):
    if num%i==0:
            print(num,"is Composite")
            break
else:
    print(num,"is Prime")
```

---

5. Write a program to print the Fibonacci series of given 'n' terms

```
num=int(input("Enter a number of terms "))
n1, n2=(-1, 1)
for i in range(1,num+1):
    n3=n1+n2
    print(n3,end=" ")
    n1, n2=n2, n3
```

---

6. Write a program to write into the text file "myfile.txt" and read the content from that text file "myfile.txt"”

SOURCE CODE

```
def WRITE():
   fh=open("myfile.txt","wt")
   data=["Twinkle Twinkle Little Star\n", "How I Wonder What You Are\n",
         "Up Above The World So High\n", "Like A Diamond In The Sky\n",
         "Twinkle Twinkle Little Star\n"]
   fh.writelines(data)
   fh.close()
WRITE()
def READ():
   fh=open("myfile.txt","rt")
   data=fh.readlines()
   for line in data:
      print(line,end="")
   fh.close()
READ()
```

---

7. Write a program to write into the text file "myfile.txt" and read the same file to display the number of vowels / consonants/ uppercase / lowercase characters in it

SOURCE CODE

```
def WRITE():
    fh=open("myfile.txt","wt")
    data=["Twinkle Twinkle Little Star\n", "How I Wonder What You Are\n",
          "Up Above The World So High\n", "Like A Diamond In The Sky\n",
          "Twinkle Twinkle Little Star\n"]
    fh.writelines(data)
    fh.close()
WRITE()
def COUNT():
    fh=open ("myfile.txt","rt")
    v,c,u,l=0,0,0,0
    data=fh.read()
    for ch in data:
        if ch in"AEIOUaeiou":
            v+=1
        if ch.isalpha() and ch not in"AEIOUaeiou":
            c+=1
        if ch.isupper():
            u+=1
        if ch.islower():
            l+=1
    fh.close()
    print("No.of vowels            :",v)
    print("No. of consonants     :",c)
    print("No. of upper cases    :",u)
    print("No. of lower cases     :",l)
COUNT()
```

---

8. Write a program to write into the text file "myfile.txt" and read the same file to display the frequency of the word (irrespective of the case) given by the user.

SOURCE CODE

```
def WRITE():
   fh=open("myfile.txt","wt")
   data=["Twinkle Twinkle Little Star\n", "How I Wonder What You Are\n",
         "Up Above The World So High\n", "Like A Diamond In The Sky\n",
         "Twinkle Twinkle Little Star\n"]
   fh.writelines(data)
   fh.close()
WRITE()
def COUNT_WORD():
    fh=open("myfile.txt","rt")
    data=fh.read().split()
    user_word=input("Enter a word to find the frequency :")
    ctr=0
    for word in data:
          if word.lower()==user_word.lower():
                ctr=ctr+1
    fh.close()
    print("Number of times",USER,"available in the file is",ctr)
COUNT_WORD()
```

---

9. Write a program to write into the text file "myfile.txt" and count the frequency of each word and display the same in the form of a dictionary.

SOURCE CODE

```
def WRITE():
   fh=open("myfile.txt","wt")
   data=["Twinkle Twinkle Little Star\n", "How I Wonder What You Are\n",
         "Up Above The World So High\n", "Like A Diamond In The Sky\n",
         "Twinkle Twinkle Little Star\n"]
   fh.writelines(data)
   fh.close()
WRITE()
```

```
def COUNT_WORDS():
    fh=open("myfile.txt","rt")
    data=fh.read().split()
    D={}
    for word in data:
        if word in D:
            D[word]=D[word]+1
        else:
            D[word]=1
    fh.close()
    print(D)
COUNT_WORDS()
```

10. Write a program to write into the text file "myfile.txt" and read the same file to display those lines starting with 'T' and its count.

SOURCE CODE

```
def WRITE():
    fh=open("myfile.txt","wt")
    data=["Twinkle Twinkle Little Star\n",  "How I Wonder What You Are\n",
          "Up Above The World So High\n", "Like A Diamond In The Sky\n",
          "Twinkle Twinkle Little Star\n"]
    fh.writelines(data)
    fh.close()
WRITE()
def TLINES():
    fh=open("myfile.txt","rt")
    data=fh.readlines()
    ctr=0
    for line in data:
      if line[0].upper()=='T':
          ctr=ctr+1
          print(line,end="")
    fh.close()
    print("No. of lines that starts with 'T' is:",ctr)
TLINES()
```

11. Write a program to create a binary file "student.dat" with rollno, name, marks in the form of a list/ tuple/ dictionary. Read the file and display the details of the students.

SOURCE CODE

```
import pickle
def WRITE():
    fh=open("student.dat","wb")
    ans='y'
    while ans in 'Yy':
        rollno=int(input("Enter the roll number : "))
        name=input("Enter the name : ")
        marks=int(input("Enter the marks : "))
        # We can pack the data using any one of the following
        data=[rollno, name, marks]              # to pack as a list
        data=(rollno, name, marks)              # to pack as a tuple
        data={'ROLLNO':rollno, 'NAME':name, 'MARKS':marks}
                                                # to pack as a dictionary
        pickle.dump(data,fh)
        ans=input("Do you want to append another record ? (Y/N)")
    fh.close()
WRITE()
def READ():
    fh=open('student.dat','rb')
    while True:
        try:
            data=pickle.load(fh)
            print(data)
        except EOFError:
            fh.close()
            break
READ()
```

---

12. Write a program to create a binary file "student.dat" with rollno, name, marks and search for the given Roll number and display the details of that student. If not found, display the appropriate message.

SOURCE CODE

```
import pickle
def WRITE():
    fh=open("student.dat","wb")
    ans='y'
    while ans in 'Yy':
            rollno=int(input("Enter the roll number : "))
            name=input("Enter the name : ")
            marks=int(input("Enter the marks : "))
            data=[rollno, name, marks]            # to pack as a list
            data=(rollno, name, marks)            # to pack as a tuple
            data={'ROLLNO':rollno, 'NAME':name, 'MARKS':marks}
                                                # as dictionary
            pickle.dump(data,fh)
            ans=input("Do you want to append another record ? (Y/N)")
fh.close()
WRITE()
def SEARCH():
    fh=open('student.dat','rb')
    RNO=int(input("Enter the Roll number to search for  "))
    while True:
            try:
                    data=pickle.load(fh)
                    if data[0]==RNO:   # if the data is of LIST / TUPLE type
                    if data['ROLLNO']==RNO:  # if data is of Dictionary type
                            print("Found data is ",data)
                            break
            except EOFError:
                    print("Sorry! The given Rollno is not available")
                    fh.close(); break
SEARCH()
```

---

13. Write a program to create a binary file "student.dat" with rollno, name, and marks. Increase 2 marks for students who have scored less than 60.

SOURCE CODE

```
import pickle
def WRITE():
    fh=open("student.dat","wb")
    ans='y'
    while ans in 'Yy':
        rollno=int(input("Enter the roll number : "))
        name=input("Enter the name : ")
        marks=int(input("Enter the marks : "))
        # We can pack the data using any one of the following
        data=[rollno, name, marks]            # to pack as a list
        data=(rollno, name, marks)            # to pack as a tuple
        data={'ROLLNO':rollno, 'NAME':name, 'MARKS':marks}
                                              # to pack as a dictionary
        pickle.dump(data,fh)
        ans=input("Do you want to append another record ? (Y/N)")
fh.close()
WRITE()
def UPDATE():
    fh=open('student.dat','rb+')
    while True:
        try:
            pos=fh.tell()
            data=pickle.load(fh)
            # if the data is of LIST type
            if data[2]<60:
                data[2]=data[2]+5

            # if the data is of TUPLE
            if data[2]<60:
                data=(data[0],data[1],data[2]+5)
```

```
                # if data is of Dictionary type
                if data['MARKS']<60:
                    data['MARKS']=data['MARKS']+5

                fh.seek(pos,0)
                pickle.dump(data,fh)
            except EOFError:
                fh.close()
                break
UPDATE()
```

---

14. Write a program to create a csv file "Employee.csv" to store Empno, Name, and Salary. Search for an Employee number and display the details. If not found, display appropriate message.

SOURCE CODE

```
import csv
def CREATE():
    fh=open("Employee.csv","wt",newline='')
    wp=csv.writer(fh)
    ans="y"
    while ans in 'Yy':
        empno=int(input("Enter the Employee number : "))
        name=input("Enter the Employee name : ")
        salary=float(input("Enter the salary : "))
        data=[empno,name,salary]
        wp.writerow(data)
        ans=input("Do you want to append another record ? (Y/N)")
    fh.close()
CREATE()
def SEARCH():
    ENO=int(input("Enter the Employee number to search  :  "))
    fh=open("Employee.csv","rt")
    rp=csv.reader(fh)
```

```
    for data in rp:
        if int(data[0])==ENO:
            print("Employee details: ", data)
            break
    else:
        print("Employee data is not found")
    fh.close()
SEARCH()
```

---

15. Write a program to create a csv file "Employee.csv" to store Empno, Name, Salary. Display the employees whose salary is more than 10000

SOURCE CODE

```
import csv
def CREATE():
    fh=open("Employee.csv","wt",newline='')
    wp=csv.writer(fh)
    ans="y"
    while ans in 'Yy':
        empno=int(input("Enter the Employee number : "))
        name=input("Enter the Employee name : ")
        salary=float(input("Enter the salary : "))
        data=[empno,name,salary]
        wp.writerow(data)
        ans=input("Do you want to append another record ? (Y/N)")
    fh.close()
CREATE()
def SEARCH():
    fh=open("Employee.csv","rt")
    rp=csv.reader(fh)
    for data in rp:
        if float(data[2])>10000:
            print(data)
    fh.close()
SEARCH()
```

16. Write a menu based program to perform the operation on stack to store the flight details (i.e) fno, fname, fare

SOURCE CODE

```
stack=[]
def push(stk, item):
   stk.append(item)
def pop():
   if stack==[]:
          print("stack is empty")
   else:
          item=stack.pop()
          print("Deleted item is:",item)
def peek():
   item=stack[-1]
   print("Peeked item is:", item)
def display():
   for i in range (len(stack)-1,-1,-1):
          print(stack[i])
#main program
print("STACK OPERATIONS")
print("1.Push")
print("2.Pop")
print("3.Peek")
print("4.Display")
while True:
          choice=int(input("Enter your choice:"))
          if choice==1:
                 fno=int(input("enter flight number:"))
                 fname=input("enter flight name:")
                 fare=float(input("enter fare:"))
                 flight=[fno,fname,fare]
                 push(stack,flight)
          elif choice==2:
                 pop()
```

```
elif choice==3:
        peek()
elif choice==4:
        display()
else:
        print("invalid choice")
        break
```

---

17. **Create table Student and insert the following records. Write the SQL queries for the given statement and also the output based on the table.**

Rollno	Name	Gender	Age	Dept	DOB	Fees
1	Arun	M	24	Computer	1997-01-10	120
2	Ankit	M	21	History	1998-03-24	200
3	Anu	F	20	Hindi	1996-12-12	300
4	Bala	M	19	NULL	1999-07-01	400
5	Charan	M	18	Hindi	1997-09-05	250
6	Deepa	F	19	History	1997-06-27	300
7	Dinesh	M	22	Computer	1997-02-25	210
8	Usha	F	23	NULL	1997-07-31	200

a) Display the unique departments.
b) List the names of female students in Hindi department.
c) List the names of the students whose name have second alphabet as 'n'
d) Delete the details of rollno 8
e) Change the fee to 170 of student whose rollno is 1 if the existing fee is less than 130.
f) Add a new column Area of datatype varchar of size 20.
g) Add the value "TRY" in the Area attribute whose department is Hindi or Computer.
h) Display the details of students whose Area is null.

QUERIES and SAMPLE OUTPUT

a) SELECT DISTINCT(dept) FROM student;

b) SELECT name FROM student
WHERE gender='F' AND dept='Hindi';

c) SELECT name FROM student
WHERE name LIKE '_N%';

---

d) DELETE FROM student
WHERE rollno=8;
SELECT * FROM student;

e) UPDATE student
SET fees =170
WHERE rollno=1 AND fees<130;
SELECT * FROM student;

f) ALTER TABLE student
ADD area VARCHAR(20);
SELECT * FROM student;

g) UPDATE student
SET area="TRY"
WHERE dept IN('Hindi','Computer');
SELECT * FROM student
WHERE dept IN('Hindi','Computer');

h) SELECT * FROM student WHERE area IS NULL;

---

18. Create Tables 'salesperson' and 'item'. Insert the records as per the data given below. Write the SQL Queries for the given statement and also the output based on the tables.

TABLE : Salesperson			
Code	Name	Salary	ITCode
1001	Tandeep Jha	60000	12
1002	Yogaraj Sinha	70000	15
1003	Tezin Jack	45000	12
1004	Anokhi Raj	50000	17
1005	Tarana Sen	55000	17

TABLE : Item		
ITCode	ItemType	Turnover
15	Stationary	3400000
17	Hoistery	6500000
12	Bakery	10090000

a) Display the code and name of all salespersons having '17' as itemtype code from the table salesperson.
b) Display all details from table salesperson in descending order of salary.
c) Display the number of salepersons dealing in each type of item.
d) Display the NAMEs of all salespersons and their corresponding ItemType
e) Display the highest salary based on type of item.

QUERIES and SAMPLE OUTPUT

a) SELECT code, name
   FROM salesperson
   WHERE itcode=17;

b) SELECT * FROM salesperson
   ORDER BY salary DESC;

c) SELECT itcode, COUNT(*)
   FROM salesperson
   GROUP BY itcode;

d) SELECT name, itemtype
   FROM salesperson S, item I
   WHERE S.itcode=I.itcode;

e) SELECT itcode, MAX(salary)
   FROM salesperson
   GROUP BY itcode;

---

19. Write program to integrate Python with MYSQL Database SCHOOL to insert the records into EMP table which has the attributes (EMPID INTEGER, ENAME VARCHAR(30), GENDER CHAR(1), SALARY FLOAT(10,2)). Display all the records from the EMP table.

SOURCE CODE

```
import mysql.connector as ms
con=ms.connect(host='localhost',user='root',password='',db='school')
if con.is_connected():
    cur=con.cursor()
    ans="Y"
    while ans in'Yy':
            eid=int(input("Enter the employee id : "))
            name=input("Enter the name of the employee : ")
            gender=input("Enter the gender of the employee : ")
            salary=int(input("Enter the salary of the employee : "))
```

```
            qry=f"INSERT INTO Emp VALUES({eid},{name},{gender},{salary})"
            cur.execute(qry)
            con.commit()
            print("Record stored successfully")
            ans=input("Do you want to add another Employee details(Y/N)?..")
    cur.execute("SELECT * FROM Emp")
    data=cur.fetchall()
    for row in data:
        print(row)
con.close()
```

---

20. Write program to integrate Python with MYSQL to search an Employee using Empid and display the record if present in the already existing table EMP. If not, display the appropriate message.

SOURCE CODE

```
import mysql.connector as ms
con=ms.connect(host='localhost',user='root',password='',db='school')
if con.is_connected():
    cur=con.cursor()
    ans="Y"
    while ans in 'Yy':
        eid=int(input("Enter the employee number to search:  "))
        qry="SELECT * FROM Employee WHERE empid={}".format(eid)
        cur.execute(qry)
        data=cur.fetchone()
        if data!=None:
            print(data)
        else:
            print("No record found")
        ans=input("Do you want to search another employee (Y/N)?..")
con.close()
```

---

21. Write a program to integrate Python with MYSQL database SCHOOL to search an Employee using Empid and update the salary of that employee if present in the already existing table EMP. If not, display the appropriate message.

SOURCE CODE

```
import mysql.connector as ms
con=ms.connect(host='localhost',user='root',password='',db='school')
if con.is_connected():
    cur=con.cursor()
    ans="Y"
    while ans in 'Yy':
        eid=int(input("enter employee number to search:"))
        qry="SELECT * FROM Employee WHERE empid={}".format(eid)
        cur.execute(qry)
        data=cur.fetchone()
        if data!= None:
            print(data)
            choice=input("Do you want to update the salary (Y/N)?..")
            if choice in ['Y','y']:
                new_sal=float(input("Enter new salary of the employee : "))
                qry=f"UPDATE Emp SET Salary={new_sal} WHERE empid={eid}"
                cur.execute(qry)
                con.commit()
                print("Employee salary updated successfully")
                qry="SELECT * FROM Emp WHERE empid={}".format(eid)
                cur.execute(qry)
                data=cur.fetchone()
                print(data)
        else:
            print("Record not found")
    ans=input("Do you want to update another employee (Y/N)?..")
con.close()
```
